

Review retrospective: *the germbij forward completion and the class-(c) arc*

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Abstract

The second review of the germbij formalisation arc¹ covers the seven files that landed after it: the forward-programme completion (PRs #104–#110 — the numerator integration, the outer tails, the generic asymptotic division, and the public `rescaledMoment_hasExpansion`) and the class-(c) semi-quasi-homogeneous arc (PRs #113–#116 — the exact quasi-homogeneous moment laws, the one-grade covariance engine, and grade recovery). The independent reviewer found *no mathematical fidelity defects in either arc*: the division recursion, the tail estimates, the exponent conventions $(-\langle q, \alpha + 1 \rangle)$ against the note’s $\ell(\alpha)$, the sign of the covariance limit, and the constant-killing equivalence all verified, independently agreeing with the session’s own hand checks. What it found instead were three boundary-honesty findings — places where a docstring or a natural reading claims slightly more than the theorems deliver — each with a documentation-level fix that the review implemented under the statement-invariance gates (fingerprint byte-identical, build green). Twenty-one proof-refactor suggestions are recorded for a dedicated simplification pass.

1 Setting

The germbij fidelity-review arc resumed at the user’s direction after the perimeter tides closed the first review’s findings. Since that review’s baseline, two programmes had landed on `main`: the forward-expansion programme’s completion — stages 5c-ii through 7, proving that rescaled posterior moments of continuous polynomial-growth observables are order- N asymptotic polynomials at 0^+ — and the scoping-consulted fragment of the note’s §7.4 item (c). Neither had been independently reviewed. The review ran as one pass with the consult split per arc, against the note’s §3 (forward direction), §7.4(c) (quasi-homogeneous recovery), the two archived architecture consults, and the upstream definition corpus. The note’s two `\leanref` anchors into `QhMomentRecovery` were verified to point at the correct lines before the consult.

2 What was reviewed

Forward completion. `Multi/NumeratorExpansion.lean` (the DCT window piece with coefficients $a_j = \int P e^{-T_2} P_j$), `Multi/NumeratorTails.lean` (both outer tails beyond all orders; the packaging `numerator_hasExpansion`), `Multi/AsymptoticDivision.lean` (the generic division lemma with the recursion $c_j = (a_j - \sum_{i=1}^j b_i c_{j-i})/b_0$), and `Multi/ForwardTheorems.lean` (the public face: `rescaledMoment_hasExpansion`, its existence form, the monomial instances, and the comparison face `momentCoeff_eq_of_isLittle0`).

¹The first covered PRs #93–#103.

Class (c). `Multi/QhMomentRecovery.lean` (exact unnormalized and normalized monomial moment laws for a quasi-homogeneous potential; the moment-ratio recovery $t^{\langle q, \alpha \rangle} \langle x^\alpha \rangle_t = M_\alpha / M_0$, all multi-indices including odd; and the single-temperature comparison form), the one-grade file `Multi/GradeComparison.lean` (the pointwise engine $|e^{-W} - 1 + W| \leq W^2$ and the one-grade difference limit $(M_2(h) - M_1(h))/h^\rho \rightarrow -\text{Cov}_P(A, Q)$), and `Multi/GradeRecovery.lean` (covariance injectivity and `grade_eq_of_normalized_rates`).

3 Fidelity findings

No mathematical defects in either arc. The reviewer re-derived the historically error-prone pieces: the division recursion’s indexing and its handling of negative b_0 (through $|g| \geq |b_0|/2$ eventually), the eventual decomposition of the numerator into window plus tails with the domain indicator collapsing on the window, the integrability supplied to the positivity route for $a_0(1) = \int e^{-T_2} > 0$, the exponent bookkeeping $-(\sum_i q_i + \sum_i q_i \alpha_i) = -\ell(\alpha)$ with the volume factor cancelling in the normalized law, the sign of the covariance limit under $V_2 - V_1 \rightarrow Q$, and the equivalence of the Lean’s $Q(0) = 0$ constant-killing with the note’s positive-weighted-degree argument. All checks converged with the session’s independent pre-triage, recorded in the review log before the responses arrived.

Three boundary-honesty findings, all of the “narrower than the prose” genre:

1. **Fixed rescaled observables (forward).** The expansion theorems hold for a fixed observable of the *rescaled* variable. The note’s fixed original-scale $\varphi(w)$ becomes the q -dependent $z \mapsto \varphi(qz)$ after rescaling, and the final observable-Taylor packaging is not in these files; the monomial instances (where $\varphi_m(qz) = q^k \cdot \text{monomialTest}$) carry the content the recovery theorems consume. The top docstring’s “the germbij note’s forward direction” read broader than that.
2. **Global exact laws vs the localized cutoff law (SQH).** The note states $\int x^\alpha e^{-tP} \chi = t^{-\ell(\alpha)} M_\alpha + O(e^{-ct})$; the Lean proves the *global* model exactly, with no cutoff and no error term — a faithful strengthening for that model, but the localized bridge is unformalised. Moreover the exact identities are junk-value-consistent, taking no integrability or positivity hypotheses, so “recovery” is substantive only when the reference moments are finite with $M_0 > 0$ (the note’s standing assumptions, not taken in the file).
3. **The generic engine vs the SQH specialization.** `GradeComparison` is the correct one-grade engine, but nothing yet constructs the correction families from weighted-homogeneous pieces, identifies the rate exponent $\rho \in \mathbb{N}$ with the cleared grade difference, or runs the finite grade induction — so `GradeRecovery`’s “reaching the designated endpoint” was optimistic; “the conditional one-grade endpoint” is accurate.

4 Simplifications applied

The three documentation-level fixes the reviewer itself recommended for the findings above — and nothing else. The `ForwardTheorems` module docstring now says “the rescaled-moment core of the germbij forward direction” and names the two non-goals (observable-Taylor packaging, Wick evaluation of the coefficient integrals); the monomial-instance docstring states the original-scale bridge $\varphi_m(qz) = q^k \cdot \text{monomialTest}$ explicitly; `QhMomentRecovery` carries the two boundary notes (junk-value consistency and the unformalised localized law); and `GradeRecovery`’s endpoint claim

is qualified. Statement invariance was checked mechanically (the public-signature fingerprint of all seven files is byte-identical before and after) and the build is green. The affirmation pass endorsed all three changes. The 21 proof-refactor suggestions (growth-normalization and Gaussian domination helpers, integrand abbreviations, the reusable eventual-denominator lemma, finite-sum linearity helpers, and the rest) are recorded verbatim in the review log as a batch for a dedicated simplification pass.

5 Disagreements and open questions

No Claude/GPT disagreements; the consults’ findings coincide with the pre-triage. Escalated to the human, as candidate future work rather than defects:

- the localized cutoff law (the $O(e^{-ct})$ bridge from the global model to the note’s χ -cut statement) — one tide, the same loss-gap mechanism as the inverse-side cutoff removal;
- the SQH instantiation of the one-grade engine (weighted-homogeneous corrections, ρ = cleared grade difference, finite induction) — the consult-designated continuation if class (c) is ever re-prioritised;
- the observable-Taylor packaging for fixed original-scale observables, if the note’s forward statement is wanted verbatim.

6 Lessons

The pattern from the first review repeats cleanly: on this seabed the failure mode is never a wrong statement — it is prose (docstrings, completion banners) claiming the neighbourhood of what was proved rather than exactly what was proved. Fidelity reviews should diff the hypothesis list and the quantifier structure against the claim sentence by default; both consults were most valuable exactly there. Also worth keeping: verifying the note’s `\leanref` line numbers against the files takes seconds and closes a whole class of anchor-drift findings before the consult runs.

7 Follow-ups

- The recorded 21-item simplification batch (review log, “Triaged findings”).
- The three candidate tides above, at the user’s discretion.
- Review the perimeter files (HessianBridge, ShiftNormalization, CutoffRemoval) once PRs #109/#112/#115 merge — they are unmerged and were deliberately out of scope here.